

Surface Areas and Volumes (Set A) — MCQ Worksheet

Mathematics • Chapter 13 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. Volume of a cube of side a is:

- (A) $6a^2$
- (C) a^3

- (B) a^2
- (D) $4a^2$

Q2. Total surface area of a cube of side a is:

- (A) $6a$
- (C) a^2

- (B) $6a^2$
- (D) $4a^2$

Q3. Volume of a cuboid with dimensions l , b , h is:

- (A) lbh
- (C) $l + b + h$

- (B) $2(lb + bh + hl)$
- (D) $l + bh$

Q4. Total surface area of a cuboid is:

- (A) $2(lb + bh + hl)$
- (C) $6lb$

- (B) lbh
- (D) $2lb$

Q5. Volume of a cylinder of radius r and height h is:

- (A) $\pi r^2 h$
- (C) $\pi r^2 + h$

- (B) $2\pi r h$
- (D) $2\pi r + h$

Q6. Curved surface area of a cylinder is:

- (A) $2\pi r h$
- (C) $2\pi r^2$

- (B) $\pi r^2 h$
- (D) $\pi r h$

Q7. Total surface area of a closed cylinder is:

- (A) $2\pi r(r + h)$
- (C) $\pi r^2 h$

- (B) $2\pi r h$
- (D) $\pi r h$

Q8. Volume of a sphere of radius r is:

- (A) $\frac{4}{3} \pi r^3$
- (C) πr^2

- (B) $4\pi r^2$
- (D) πr^3

Q9. Surface area of a sphere of radius r is:

- (A) $4\pi r^2$
- (C) $2\pi r^2$

- (B) $\frac{4}{3} \pi r^3$
- (D) πr^2

Q10. Surface area of a hemisphere (curved only) is:

- (A) $2\pi r^2$
- (C) πr^2

- (B) $4\pi r^2$
- (D) $3\pi r^2$

Q11. Total surface area of a closed hemisphere is:

- (A) $2\pi r^2$
- (C) $4\pi r^2$

- (B) $3\pi r^2$
- (D) πr^2

Q12. Volume of a hemisphere is:

- (A) $\frac{2}{3} \pi r^3$
- (C) πr^3

- (B) $\frac{4}{3} \pi r^3$
- (D) $2\pi r^3$

Q13. Volume of a cone of radius r and height h is:

- (A) $\frac{1}{3} \pi r^2 h$
- (C) $\frac{1}{2} \pi r^2 h$

- (B) $\pi r^2 h$
- (D) $2\pi r^2 h$

Q14. Curved surface area of a cone (slant height l) is:

- (A) $\pi r l$
- (C) $2\pi r l$

- (B) $\pi r^2 l$
- (D) $\pi r h$

Q15. Total surface area of a cone is:

- (A) $\pi r(r + l)$
(C) $\pi r^2 + 2\pi rl$

- (B) πrh
(D) $\pi r^2 + \pi rl$

Q16. Slant height of a cone with radius r , height h is:

- (A) $\sqrt{r^2 + h^2}$
(C) $r^2 + h^2$

- (B) $r + h$
(D) $\sqrt{r^2 - h^2}$

Q17. A cone, hemisphere and cylinder stand on equal bases and have equal heights. Ratio of their volumes is:

- (A) 1:2:3
(C) 2:3:1

- (B) 1:1:1
(D) 3:2:1

Q18. If a cylinder and cone have same base and height, ratio of their volumes is:

- (A) 1:1
(C) 1:3

- (B) 3:1
(D) 2:1

Q19. A solid of dimension $4 \times 3 \times 2$ cm is melted into a cube. Side of cube is:

- (A) $\sqrt{24}$ cm
(C) 6 cm

- (B) $\sqrt[3]{24}$ cm
(D) 12 cm

Q20. A right circular cylinder has radius 7 cm and height 10 cm. Its volume ($\pi = 22/7$) is:

- (A) 1540 cm^3
(C) 440 cm^3

- (B) 154 cm^3
(D) 2200 cm^3

Q21. A cylinder of radius 5 cm and height 14 cm. Curved surface area ($\pi = 22/7$):

- (A) 440 cm^2
(C) 154 cm^2

- (B) 220 cm^2
(D) 880 cm^2

Q22. Volume of a sphere with radius 7 cm ($\pi = 22/7$) is:

- (A) 1437.33 cm^3
(C) 1372 cm^3

- (B) 4188 cm^3
(D) 1078 cm^3

Q23. A toy is in the shape of a cone surmounted on a hemisphere. Its total volume is:

- (A) $\frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$
(C) $\pi r^2 h + \pi r^2$

- (B) $\pi r^2 h + \pi r^2 h$
(D) $2\pi rh + \pi r^2$

Q24. If a cone has its height doubled keeping the radius same, its volume:

- (A) Stays the same
(C) Triples

- (B) Doubles
(D) Halves

Q25. If radius of a sphere is doubled, its surface area becomes:

- (A) 2 times
(C) 4 times

- (B) 3 times
(D) 8 times